

HILBERT SCHEME

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1. INTRODUCTION

To understand the geometry of a given algebraic variety X we often need to understand some of its subschemes. For example, if $X = \mathbb{P}^n$, then all of its smooth degree d hypersurfaces are diffeomorphic. When studying subschemes one quickly sees that subschemes comes in families, e.g.,

$$V(xy - tz^2) \subseteq \mathbb{P}_{x,y,z}^2 \times \mathbb{A}_t^1$$

defines a family of conics in $\mathbb{P}^2$. Hence, to understand subschemes we will study them in families. We achieve this via the *Hilbert functor*. Once we introduce the Hilbert functor we will show that the Hilbert functor is representable by a projective scheme, which we call the *Hilbert scheme*. In what proceeds we work over $k = \bar{k}$. Every construction extends to the relative setting, but we work over k for simplicity.

2. HILBERT FUNCTOR

Let X/k be a projective variety and let $\mathcal{O}_X(1)$ be a fixed very ample line bundle. Before defining the Hilbert functor we will introduce the *Hilbert polynomial*.

Definition-Proposition 2.1.